

Assignment 5: Detailed Final Project Report

Multi-Agent Systems Engineering (1-3520104-20)
Due Date: 22/07/2026 | Weight: 45% of Final Grade

1. Objective

The Final Project Report is the definitive technical and scientific document of your MAS project. It synthesizes the entire engineering lifecycle, from problem definition to final simulation results.

2. Mandatory Table of Contents

1. **Chapter 1: Introduction & Executive Summary:** Project vision, problem domain, and a summary of key achievements.
2. **Chapter 2: Final Requirements & Design Recap:** Summary of agent roles, environment taxonomy, and a list of justified design changes from the original SDD.
3. **Chapter 3: Technical Implementation:** Development stack, software architecture, and the mechanics of the simulation engine.
4. **Chapter 4: Agent Logic & Action Strategies:** Detailed pseudo-code or flowcharts of the decision-making process (e.g., Nash Equilibrium, Reinforcement Learning, or BDI Logic).
5. **Chapter 5: Simulator & Visualization Platform:** User manual for the simulator and explanation of the real-time visualization dashboard and data logging.
6. **Chapter 6: Results & Validation Analysis:** Empirical evidence of system success, including Utility and Social Welfare charts compared to baseline strategies.
7. **Chapter 7: Social, Ethical & Future Work:** Discussion of autonomy ethics in your domain and potential future improvements.

3. Deliverables

- 1. Full Final Report (PDF format).
- 2. Project Source Code & Simulation Executable. Link in the PDF.
- 3. Video Demo (3-5 minutes showing the simulator in action). Link in the PDF.